

Milestone 2 Report

Smart MCQ Solver Challenge

Course: BSDA2001P – Introduction to Deep Learning and Generative AI

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Objective

This milestone focuses on transitioning from classical NLP techniques to transformer-based methods. It introduces the Hugging Face datasets and transformers libraries, BERT architecture, attention mechanisms, context-aware sentence embeddings, zero-shot classification, and generative language models.

Work Performed

- Loaded train.csv using the Hugging Face datasets library.
- Created a combined_text column using Dataset.map().
- Initialized the bert-base-uncased tokenizer and explored tokenizer properties.
- Generated tokenized tensors with padding, truncation and PyTorch tensors.
- Extracted contextual embeddings and CLS representations from BERT.
- Visualized attention behaviour using output_attentions=True.
- Generated sentence embeddings using sentence-transformers/all-MiniLM-L6-v2.
- Built TF-IDF and MiniLM ranking pipelines and evaluated them using MAP@3.
- Performed zero-shot classification using facebook/bart-large-mnli.
- Generated answers using google/flan-t5-small (implemented via AutoModelForSeq2SeqLM because the installed Transformers 5.0 environment no longer supports the text2text-generation pipeline task).

Key Results

Experiment	Result
Combined text length (row 51)	614
BERT vocabulary size	30522
[SEP] token ID	102
Tokenized input_ids shape	torch.Size([2000, 128])
Attention head size	64
last_hidden_state shape	torch.Size([1, 31, 768])
Sum of first 5 CLS values	-1.2001
CLS → fusion attention weight	0.1025
MiniLM cosine similarity	0.7658
TF-IDF MAP@3	0.3448

MiniLM MAP@3	0.4231
Improved questions over TF-IDF	513
Zero-shot top probability	0.4575
Softmax vs Sigmoid difference	0.9995
FLAN-T5 generated output	B

Observations

- MiniLM achieved a higher MAP@3 score (0.4231) than the TF-IDF baseline (0.3448), demonstrating the benefit of contextual embeddings.
- Transformer models capture semantic meaning better than lexical matching alone.
- Attention weights provide interpretability regarding token interactions.
- Zero-shot classification enables inference without task-specific fine-tuning.
- Generative language models can answer questions directly from prompts.